PAPER ANALYSIS / PARTIAL REPLICATION

Two Types of Attention, Two Paths to Market Returns

Market Returns and a Tale of Two Types of Attention

Research logic and a partial replication of Tables 2–4

Da, Hua, Hung & Peng

The presentation follows one chain of questions

How can attention to individual stocks affect the whole market?

01 Mechanism

Retail buying pressure versus institutional information processing.

02 Measurement

Firm-level attention aggregated with market-value weights.

03 Tests

Future returns, market states, and announcement windows.

Table 2: baseline → Table 3: states → Table 4: news

The aggregation problem

Firm-level evidence does not automatically imply market predictability

At the stock level

Attention can shift demand and prices for an individual stock.

At the market level

Effects across stocks may arrive at different times, offset one another, or receive different value weights.

Attention effect for one stock $\not\Rightarrow$ predictable market return

Three ways a stock-level effect can disappear

A market result requires more than many firm-level results

Idiosyncratic shocks

Firm-specific attention can average out.

Small-stock concentration

A strong effect on small firms can carry little value-weighted impact.

Asynchronous timing

Reversal in one stock can coincide with fresh pressure in another.

Design implication: Aggregate attention from individual firms, then test market returns directly.

Attention is a scarce resource

The same word can describe different choices

Attention captured by salience

A striking price move or headline can pull an investor toward a stock.

Deliberate allocation

An investor can devote scarce research capacity to an anticipated event.

	Likely behavior	Return implication
Retail	Notice, then buy; buying is more attention-sensitive	Temporary pressure may reverse
Institutional	Read and analyze before an event	News resolution may earn a premium

Retail channel I: why attention may generate net buying

Attention enters the two sides of a trade differently

Buying

The choice set is large; noticed stocks are more likely to become candidates.

Selling

Investors usually choose from stocks already held; attention has a smaller screening role.

More attention ⟶ relatively more buying

Takeaway This is a behavioral hypothesis: attention affects demand before it affects price.

Retail channel II: from buying pressure to reversal

Suppose buying conveys little new fundamental information



On the attention day

Buying can push prices above fundamentals.

Over the next week

As pressure fades, subsequent returns may turn negative.

Retail channel III: when should reversal be stronger?

Price pressure persists when arbitrage is difficult

High VIX

Liquidity providers demand more compensation for risk.

Wide spreads

A given order imbalance creates more price impact.

High lending fees

Short-sale trades that correct prices cost more.

Prediction: ARA has a more negative future-return coefficient in these states.

Institutional channel: why attention rises before news

Research capacity can be allocated ahead of an event



Key condition

Ahead of major news, attention may signal that an event matters to many stocks.

Full sample

Without an event condition, the positive association can be diluted.

Takeaway The conditional premium can reflect risk-premium realization or resolution of uncertainty.

Two attention measures, two testable predictions

Different mechanisms require different tests

	Retail ARA	Institutional AIA
Baseline	Lower return over the following week	Weak unconditional relation
Conditional effect	More negative when market frictions rise	More positive before major news
Interpretation	Buying pressure followed by correction	Information processing and uncertainty resolution

The core tables ask, in order: **Does it predict? When is it stronger? What mechanism fits?**

Sample and data: from stocks to the market

The published study's common framework

Stock universe

US common shares on NYSE, AMEX, NASDAQ, and NYSE Arca, aligned with firm market value.

Coverage

ARA: July 2004–December 2019.
AIA: available from February 2010.

Retail input

Google search volume

Institutional input

Bloomberg readership

Market return

CRSP value-weighted index

Sample construction matters for the market test

The paper does not use a current index membership list

Eligibility

CRSP common-share codes 10 and 11; stocks closing below $5 are excluded.

Coverage shift

Google attention begins in 2004; Bloomberg readership begins in 2010.

Different ARA and AIA samples affect precision and complicate direct coefficient comparisons.

ARA begins with abnormal firm-level search

Google Search Volume Index / ASVI

$$ASVI_{i,t} = \frac{SVI_{i,t}}{\text{median}(SVI_{i,t-1:t-6m})} - 1$$

Why abnormal?

Compare a stock with its own six-month history, rather than with other stocks' raw search volumes.

What is observed?

Active searches for a stock's name or ticker, rather than news volume or trading.

Three details in constructing Google attention

Measurement choices affect the interpretation

Ticker ambiguity

Generic tickers are manually screened.

Rolling baseline

The benchmark must use information available before day *t*.

Data vintages

Daily SVI is a 0–100 relative score; vintages are cross-checked.

These choices determine whether aggregated searches reliably represent retail attention.

AIA measures attention through Bloomberg readership

Turn firm-level reading intensity into a high-attention indicator

$$HighInstAttention_{i,t} = \mathbb{1}(DMR_{i,t} \geq 3)$$

DMR

Maximum hourly Bloomberg readership, relative to the stock's prior 30 days.

High attention

Scores 3 or 4 exceed roughly the 94th percentile of past activity.

From firms to the market: one aggregation rule

Larger firms receive greater weight

$$ARA_t = \sum_i w_{i,t} ASVI_{i,t} \qquad AIA_t = \sum_i w_{i,t} HighInstAttention_{i,t}$$

How to read ARA

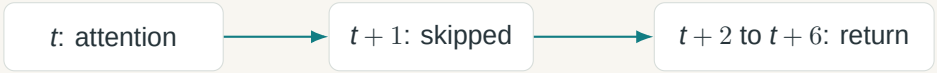
Market-wide abnormal search intensity.

How to read AIA

The value-weighted share of stocks receiving high institutional attention.

Timing: attention first, returns later

The main outcome is the cumulative return on trading days $t+2$ through $t+6$



Why skip a day?

Calendar-day attention overlaps part of the next close-to-close return.

Why a five-day window?

To capture a short correction or returns around information release.

The clock mismatch explains the skipped day

The timing choice follows how the source data define a day

Google ARA

The midnight-to-midnight GMT search window overlaps about four hours of the $t+1$ market return.

Bloomberg AIA

Calendar-day readership overlaps about eight hours of the $t+1$ return.

Using $R_{t+2:t+6}$ keeps these overlapping hours out of the headline predictive test.

Regression: what else is controlled?

Predict future market returns with today's attention

$$R_{t+2:t+6} = \alpha + \beta Attention_t + \gamma X_t + \varepsilon_t$$

Macro and sentiment

BW, TMS, DFY, ΔADS, ΔEPU

News and trading

Abnormal news volume and turnover

Market dynamics

VIX, turnover, returns, and up to four lags

Table 1: the two attention measures differ

Descriptive statistics in the published study

Measure	Trading days	Mean	Std. dev.	Persistence ρ
ARA	3,903	0.065	0.054	0.77
AIA	2,431	0.254	0.100	0.57

Common component

ARA and AIA have a correlation of about 0.282; they are not independent.

Important differences

Their volatility, persistence, and economic meanings are distinct.

Aggregation reveals a persistent common component

Market attention outlasts the average firm-specific shock

Autocorrelation	Aggregate measure	Average firm-level measure
Retail attention	0.77	0.41
Institutional attention	0.57	0.24

Reading

Shared shocks persist after firm-specific noise is averaged out.

Caution

Attention is still less persistent than VIX and yield spreads.

Dynamics: which attention measure tends to lead?

The paper studies lead–lag relations with a VAR

After an AIA shock

ARA's cumulative response reaches about 0.219 by day nine and remains significant.

After an ARA shock

AIA's initial response is about 0.079 and is not significant.

Institutional attention $\longrightarrow$ **Retail attention**

Takeaway Impulse responses describe dynamic association; they do not establish causation.

Table 2: establish the baseline first

The same return windows are tested with ARA and AIA separately

Panel A: ARA

Track each day from $t+1$ to $t+6$, then the cumulative return from $t+2$ to $t+6$.

Panel B: AIA

Use the same timing; add ARA in one specification to distinguish the two signals.

Establish the **average predictive relation** before conditioning on states and events.

Table 2A: ARA predicts subsequent daily returns

Published results / retail attention

Return day	$t+1$	$t+2$	$t+6$	Pattern
ARA coefficient	-0.814^{**}	-0.902^{**}	-0.805^{***}	Negative on several future days
NW *t*	-2.24	-2.30	-2.64	Persists beyond day one

Direction

Higher ARA is followed by lower daily market returns.

Timing

The main test starts at $t+2$ to reduce ambiguity around measurement and nearby returns.

Table 2A: the economic magnitude over one week

Published result / controlled $t+2:t+6$ specification

$$\beta = -4.137^{***}$$

Newey–West $t = -2.86$

One typical change

A one-standard-deviation rise in ARA corresponds to about 22.34 bp lower market return over the following week.

Mechanism link

Consistent with retail buying pressure followed by a correction.

Table 2B: AIA's daily signal is different

Published results / institutional attention

Return day	$t+1$	$t+2$	$t+6$	Interpretation
AIA coefficient	0.396^{*}	0.240	0.053	Most daily results are not significant
NW *t*	1.75	1.18	0.32	Unlike ARA's timing pattern

The positive signal from institutional attention may require a **major information event**, rather than an arbitrary trading day.

Table 2B: weak full-sample weekly predictability

Published result / controlled $t+2:t+6$ specification

AIA alone

Coefficient 0.927; NW $t = 1.52$, below conventional significance thresholds.

Controlling for ARA

AIA coefficient 1.089^{*}; NW $t = 1.72$, still relatively weak evidence.

Next question: Is AIA informative specifically before announcements?

Partial replication: retained and substituted inputs

The feasible specification for Tables 2–4

Element	Published study	Replication exercise
Retail attention	Google ARA	Authors' released ARA
Institutional attention	Bloomberg AIA	Abnormal index-options volume proxy
Market return	CRSP broad market	S&P 500 proxy
Sample	Separate ARA and AIA coverage	2010–19; 2,437 common trading days
Abnormal news volume	Included control	Unavailable; omitted

Table 2 replication: ARA retains the negative sign

Controlled return over the following week

	Published study	Replication
Coefficient	-4.137^{***}	-1.363
NW30 *t*	-2.86	-1.10
One-SD effect	-22.34 bp	-7.28 bp
Trading days	3,903	2,437

Reading: The sign matches, but the estimated effect is smaller and statistically inconclusive in the replication.

Table 2 replication: the AIA proxy is insignificant

Controlled return over the following week

	Original Bloomberg AIA	Options-volume proxy
Coefficient	0.927	0.197
NW30 *t*	1.52	1.10
Trading days	2,431	2,437

Similarity

Neither weekly full-sample estimate provides strong evidence.

Boundary

Readership and options trading are different measures; coefficient sizes are not directly comparable.

Table 2 in one slide: a baseline and a condition

The next tests examine mechanisms

ARA

Significantly negative in the paper; negative but insignificant in the replication. Next: do market frictions amplify it?

AIA

Limited full-sample evidence. Next: does the signal appear before announcements?

Table 3: market states

Table 4: news events

Table 3: bring market frictions into the test

If price pressure is the channel, market states should change its strength

High VIX

Liquidity providers are more sensitive to risk.

Wide effective spreads

Trading pressure is harder to absorb.

High stock-lending fees

Short-sale correction becomes more costly.

Split the sample by state, compare ARA coefficients, and test the high-minus-low difference.

Table 3: stronger reversal when VIX is high

Published result / VIX split

ARA	High VIX	Low VIX	High minus low	Interpretation
Coefficient	-5.167	-1.170	-3.997	Negative difference
White *t*	-2.92^{***}	-1.23	-1.99^{**}	Stronger in high VIX
One-SD effect	-27.96 bp	-6.29 bp		

Takeaway In the published results, retail attention predicts a larger reversal during high-volatility states.

Table 3: two other frictions point in the same direction

Published results / these splits are outside the replication

Effective spreads

ARA coefficient: -6.303 in the high-spread group versus -0.667 in the low-spread group.

Stock-lending fees

ARA coefficient: -17.968 in the high-fee group versus -4.000 in the low-fee group.

All three splits fit the idea that stronger frictions slow correction of short-lived buying pressure.

Table 3 replication: ARA's VIX difference is uncertain

Only the VIX split was reconstructed / replication uses NW30 inference

ARA	High VIX	Low VIX	High minus low	NW30 *t*
Replication coefficient	-2.192	-0.692	-1.500	$-1.11,\ -0.29,\ -0.43$
Published coefficient	-5.167	-1.170	-3.997	Paper also reports White *t*

Point estimates

The high-VIX coefficient is more negative, as in the paper.

Inference

The high-minus-low difference is not significant in the replication.

Table 3 replication: the AIA proxy across VIX states

The inference method matters

AIA	High VIX	Low VIX	High minus low	Difference *t*
Published study	1.480	-0.200	1.680	White 1.96^{**}
Options-volume proxy	0.326	-0.166	0.491	White 1.81^{*}
Replication NW30				1.43

Takeaway The proxy offers weak 10% evidence with White errors, but not after allowing for time dependence.

Table 3 synthesis: separate direction from mechanism

How to read the state tests

Published pattern

ARA is more negative under high VIX, wide spreads, and high lending fees.

Replication pattern

The high-VIX point estimate is more negative, but the difference is not robustly significant.

An additional TED-spread analysis concerns **funding stress**; it is not a measure of stock-lending fees and cannot replace that test.

Table 4: attention before news

A weak full-sample effect can hide a conditional relation



Prediction

AIA's positive coefficient is larger if a major announcement occurs in the future-return window.

Comparison

Days when that future window has no announcement.

Two explanations for the AIA news-window premium

The paper treats these channels as compatible, not mutually exclusive

Risk-premium timing

Announcements carrying systematic information can concentrate expected equity premia near release.

Uncertainty resolution

High institutional attention can signal elevated, anticipated market uncertainty before news.

Table 4 shows a conditional predictive association; it does not by itself identify which channel dominates.

Table 4: what counts as a news window?

The announcement must occur between $t+2$ and $t+6$

Macro News

FOMC meetings, nonfarm payrolls, or PPI releases.

All News

Macro news or days in the top 5% of announcing firms' market-capitalization share.

The split is based on **whether an announcement occurs**, not whether its content is good or bad.

Institutional attention rises before scheduled news

The event split is also reflected in the attention series itself

Mean AIA	Before major news	Other days
Published sample	0.266	0.241

Evidence

The difference is statistically strong ($t = 5.84$).

Interpretation

Institutions allocate more attention before events that can affect many firms.

Table 4A: AIA's positive signal appears around news

Published result / Macro News

AIA	News	No news	News minus no news	White *t*: news; diff.
Coefficient	2.076	-0.375	2.451	$3.20^{***};\ 2.78^{***}$
One-SD effect	$+20.82$ bp	-3.74 bp		

Key comparison

The news-window coefficient is positive, and the between-window difference is significant.

Mechanism link

Institutional attention may reflect deliberate analysis before news resolves uncertainty.

Does AIA simply capture informed front-running?

The paper checks returns before scheduled macro announcements

Pre-event window

Correlations between AIA and market returns from $t-4$ to t are small: 0.011 to 0.038.

Interpretation

No notable pre-release return pattern appears; this is consistent with uncertainty being anticipated.

The check narrows one alternative explanation, but it is not a complete test of private information trading.

Table 4B: ARA’s news-window difference

Published result / Macro News

ARA	News	No news	News minus no news	White *t*
Coefficient	-2.505	-5.225	2.720	$-1.69^{*},\ -4.50^{***},\ 1.44$

Different from AIA

ARA remains mainly negative; its between-window difference is insignificant.

Interpretation

Retail pressure and institutional pre-news analysis imply different return paths.

How was the Macro News indicator reconstructed?

Preserve the event definition while recognizing differences in implementation

Real dates

Use official FOMC, payroll, and PPI announcement calendars.

Future window

Mark whether an event falls on trading days $t + 2$ through $t + 6$.

Regression sample

1,324 news-window days and 1,113 non-news days.

Event selection and holiday treatment may differ from the paper. AIA remains an options-volume proxy, and abnormal news volume is omitted as a control.

Table 4 replication: no significant AIA news effect

Macro News / index-options-volume proxy

	News	No news	News minus no news	Difference *t*
Original AIA	2.076	-0.375	2.451	White 2.78^{***}
Options-volume proxy	0.280	0.209	0.071	NW30 0.22

Point estimate

The coefficient on news-window days is positive.

Decisive contrast

The difference from non-news days is near zero and insignificant.

Replication: ARA's news-window difference

Macro News / retail attention

	News	No news	News minus no news	Difference *t*
Published study	-2.505	-5.225	2.720	White 1.44
Replication	-2.333	0.322	-2.655	NW30 -1.17

Common feature

Neither between-window difference is significant.

Important difference

Their signs differ, so the results are not uniformly aligned.

Table 4 synthesis: the window difference is central

Do not read one positive coefficient in isolation

Published claim

AIA's positive predictability rises significantly before major announcements.

Replication evidence

The proxy's news-minus-no-news difference is insignificant.

The replication does not refute the original AIA result: it observes options trading, while the paper observes Bloomberg readership.

Section 2.4: does retail attention accompany buying?

Published trading evidence

Retail order imbalance

ARA is positively associated with measures of net retail buying.

Mutual-fund flows

ARA is also positively associated with equity-fund inflows.

Contemporaneous trading outcome	Coefficient	NW *t*
ARA $\rightarrow$ ROIB1	3.555^{**}	2.25
ARA $\rightarrow$ fund flow	0.187^{**}	2.42

Section 2.4: retail buying on down-market days

Published trading evidence

Negative-return days

The association between ARA and net retail buying is stronger, consistent with contrarian purchases.

Extreme down days

The buying measures remain positive when market returns are in the bottom 5% of the sample.

Buying even after bad news can create short-lived support for prices followed by a correction.

Out-of-sample forecast design

The paper moves from fitted relationships to implementable predictions

Rolling estimation

Estimate a univariate attention regression on a 500-day window.

Benchmark

Use the historical mean return from the same window.

Evaluation

Compare forecast errors for the $t + 2 : t + 6$ return.

ARA test starts

August 2006, after the initial training period.

AIA test starts

March 2012, after Bloomberg coverage allows training.

Section 2.5: does predictability survive out of sample?

Published Table 5

Out-of-sample R^2	Full sample	Relevant state	Comparison	Pattern
ARA	$1.49\%^{**}$	High VIX $2.70\%^{***}$	Low VIX -0.39%	State dependence
AIA	-0.36%	Macro News $1.19\%^{*}$	No news -0.39%	Event dependence

Takeaway The out-of-sample pattern mirrors the paper's central distinction: frictions for retail, news for institutions.

Does forecasting improve allocation?

The paper adds an economic, rather than only statistical, test

Portfolio rule

A mean–variance investor splits wealth between the market and Treasury bills using the return forecast.

Assumptions

Risk aversion is 3; the reported strategy includes 50 bp transaction costs.

Annual certainty-equivalent gain	Full sample	Strongest relevant state
ARA	2.26%	High VIX 2.76%
AIA	-0.74%	Macro News 1.83%

Section 2.6: are the results robust to other choices?

Published Table 6

Time and sample

Exclude the crisis or December; add weekday effects.

Inference

Change standard errors, add lagged attention, or smooth attention.

Measures

Use S&P 500 returns or alternative value weights.

ARA

The coefficient remains negative in most checks.

AIA around news

The conditional coefficient remains positive in most checks.

Alternative aggregation: what if mega-caps dominate?

Published Table 6, Panel B

Reweighting

Use capped market-value weights, equal weights, principal components, or partial least squares.

Result

PLS attention is stronger: ARA has OOS $R^2 = 2.14\%$; AIA in All News windows has $R^2 = 1.36\%$.

The main effects survive substantial reductions in mega-cap weight; they are not driven by a few giant firms.

Why bottom-up attention beats top-down proxies

Published Table 6, Panel C

Retail

Searches for market-index terms do not predict returns reliably out of sample. Retail investors often hold and trade individual stocks.

Institutional

Readership of SPY alone is weaker than the share of individual stocks receiving high Bloomberg attention.

Bottom-up AIA also signals how broadly an upcoming event matters across the stock universe.

Cross-sectional design: test exposure to each channel

Published Table 7 asks where market-level predictability should be strongest

Liquidity portfolios

Sort stocks by past-month Amihud illiquidity; compare the most liquid and least liquid quintiles.

Beta portfolios

Estimate CAPM beta from rolling monthly returns; compare low- and high-beta quintiles around news.

These sorts test price-impact sensitivity for ARA and systematic-risk exposure for AIA.

Section 2.7: which portfolios bear more of the effect?

Published Table 7

Liquidity sorting

ARA's negative predictability is stronger for illiquid stocks.

Full sample: -3.443 for the most liquid portfolio versus -5.675 for the least liquid.

Beta sorting

Before major news, AIA's positive predictability is stronger for high-beta stocks.

Macro News: 1.284 for low beta versus 3.512 for high beta.

The cross-section connects the two channels to **liquidity exposure** and **market-risk exposure**.

Distraction instrument: what variation is used?

The paper seeks attention shocks unrelated to stock fundamentals

Construct Dist

Take the top 10% of news-pressure days within each year, based on sensational broadcast-news coverage.

Restrict the sample

Exclude scheduled macro news, extreme market moves, and the crisis period to limit direct market-news channels.

The identifying assumption is that these distractions affect future returns through retail attention rather than another channel.

Section 3.1: news distraction as an attention shock

Published Table 8

First stage

On high-news-pressure distraction days, ARA falls: coefficient -0.014^{***} and first-stage $F = 19.673$.

Second stage

Instrumented ARA predicts future returns with coefficient -23.737; NW $t = -1.87^{*}$.

The instrument is weak for AIA ($F = 0.687$), so it offers no strong causal evidence for the institutional channel.

The clustered-earnings puzzle comes before the release

Why is an after-hours announcement associated with an earlier market premium?

Event definition

In January, April, July, and October, identify the four days with the greatest announcing-firm market capitalization.

Timing puzzle

For clustered after-hours releases, the market earns about 17.16 bp on the event day, before the news arrives.

Split these days by ARA: if retail buying pressure matters, the early premium should be concentrated on high-ARA days.

Section 3.2: clustered earnings and return timing

Published Figure 3 and Table 9

High-ARA after-hours earnings days

The same-day coefficient is $+33.478$ bp and significant; the next-day coefficient is -9.794 bp and insignificant.

Low-ARA after-hours earnings days

The same-day coefficient is $+9.396$ bp and insignificant; the next-day coefficient is $+26.455$ bp and significant.

Retail buying before an announcement can change which day captures the return premium.

Final synthesis: the paper and the replication

Return to the two paths

Retail in the paper

ARA → buying pressure → reversal; stronger under market frictions.

Institutional in the paper

AIA → pre-news analysis → positive news-window returns.

Partial replication

Some point-estimate signs align, but robust significance in Tables 2–4 is not recovered.

Who pays attention shapes the mechanism; when they pay attention determines when it appears in returns.